Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. Which of the following are considered potential security vulnerabilities that could be exploited within AI systems? (Select all that apply.) 1 / 1 point

- [ ] Hyperparameter tuning
- [ ] Feature scaling
- [x] Model inversion

Nice work

Correct! Model inversion attacks aim to deduce sensitive information from an AI model by exploiting its responses.

- [x] Data poisoning

Nice work

Correct! Data poisoning refers to the manipulation of training data to corrupt the learning process, resulting in erroneous AI model outputs.

- [x] Adversarial examples

Nice work

Correct! Adversarial examples are maliciously crafted inputs that cause an AI model to make incorrect predictions or classifications.

2. Which of the following are commonly employed techniques aimed at securing AI models against attacks and enhancing their robustness? (Select all that apply.) 1 / 1 point

- [x] Federated learning

Nice work

Correct! Federated learning allows AI models to be trained across decentralized devices without sharing raw data, ensuring privacy.

- [x] Homomorphic encryption

Nice work

Correct! Homomorphic encryption enables computation on encrypted data without decrypting it, thereby protecting the data during processing.

- [x] Differential privacy

Nice work

Correct! Differential privacy introduces noise into the data to safeguard individual privacy while allowing aggregate analysis.

- [ ] Data sharding
- [ ] Gradient descent

3. Which cryptographic technique is most commonly used to verify the integrity of data and ensure that it has not been altered within AI systems? 1 / 1 point

- (•) Hash functions
- ( ) Symmetric encryption
- ( ) Steganography
- ( ) Asymmetric encryption

Nice work

Correct! Hash functions produce a unique, fixed-length output for any input data, allowing verification that the data remains unaltered.

4. Which evaluation metric is more suitable for assessing the performance of a fraud detection algorithm, especially when dealing with highly imbalanced datasets? 1 / 1 point

- (•) ROC AUC
- ( ) Accuracy
- ( ) Precision
- ( ) Recall

Nice work

Correct! The ROC AUC metric evaluates performance across various classification thresholds, making it ideal for imbalanced datasets where the positive class is rare.

5. Which of the following steps are essential when evaluating a fraud prevention algorithm using metrics such as F1 score and ROC AUC? (Select all that apply.) 1 / 1 point

- [x] Calculating the confusion matrix

Nice work

Correct! The confusion matrix provides detailed insight into the performance of the model, enabling calculation of F1 score and ROC AUC.

- [x] Splitting the dataset into training and testing sets

Nice work

Correct! Partitioning the data into training and testing sets is crucial to assess how well the model generalizes to unseen data.

- [ ] Clustering transactions by type before evaluation
- [x] Using cross-validation

Nice work

Correct! Cross-validation enhances the robustness of the model's performance evaluation by testing it on multiple data subsets.

- [ ] Normalizing the transaction amounts

6. When developing a credit card fraud detection system using IBM Watson, which of the following is a critical step for improving the algorithm's detection accuracy? 1 / 1 point

- (•) Training the model with labeled fraudulent and non-fraudulent transactions
- ( ) Reducing the dimensionality of the transaction features
- ( ) Using data augmentation techniques to create synthetic transactions
- ( ) Applying ensemble methods to combine multiple models

Nice work

Correct! Providing labeled transaction data enables the AI model to learn to distinguish between genuine and fraudulent transactions, thereby improving accuracy.

7. Which of the following factors are crucial for understanding and addressing credit card fraud within financial institutions? (Select all that apply.) 1 / 1 point

- [ ] Training staff in cybersecurity
- [x] Historical transaction patterns

Nice work

Correct! Analyzing historical transaction data helps uncover unusual patterns that may indicate fraudulent activity.

- [x] Customer spending behavior

Nice work

Correct! Monitoring deviations from typical spending behavior can provide insights into potentially fraudulent transactions.

- [ ] Physical security measures at ATMs
- [x] Types of fraud techniques used

Nice work

Correct! Familiarity with different fraud techniques is essential for developing preventive measures and effective detection strategies.

8. What are the major implications of credit card fraud for financial institutions, and how do they affect the institution's overall operations? (Select all that apply.) 1 / 1 point

- [x] Financial losses due to fraudulent transactions

Nice work

Correct! Fraudulent transactions lead to substantial financial losses, directly affecting the profitability of the institution.

- [x] Damage to the institution's reputation

Nice work

Correct! Fraud can erode trust and significantly damage the reputation of a financial institution, leading to customer attrition.

- [x] Legal penalties for failing to prevent fraud

Nice work

Correct! Institutions may face legal consequences if they do not implement adequate measures to prevent fraud.

- [x] Increased operational costs for fraud prevention

Nice work

Correct! Implementing comprehensive fraud prevention measures often results in higher operational costs for the institution.

- [ ] Improved customer satisfaction

9. What is one of the primary challenges associated with using data-driven predictive models for fraud prevention in rapidly evolving fraud landscapes? 1 / 1 point

- (•) Need for continuous data updates
- ( ) High dependence on expert input
- ( ) Inflexibility to adapt to new fraud patterns
- ( ) Limited ability to handle large datasets

Nice work

Correct! Data-driven models must be frequently updated with new data to remain effective against emerging fraud tactics.

10. Which of the following characteristics best describes rule-driven fraud prevention models in contrast to more modern data-driven approaches? 1 / 1 point

- (•) Depend on predefined rules created by experts
- ( ) Automatically update to new types of fraud
- ( ) Utilize machine learning to detect complex patterns
- ( ) Require large datasets for effective operation

Nice work

Correct! Rule-driven models rely on a fixed set of rules established by domain experts for fraud detection, whereas data-driven models learn from data.